

H2R Sports — Database Structure

MongoDB · Mongoose | Database: h2r-sports | E-commerce reference schema (catalog, users, checkout, orders, marketing, store billing)

9 collections

Paid orders only after verify

Pending checkout TTL 24h

1. Overview

#	Collection	Purpose
1	users	Customers & admins + saved addresses
2	products	Product catalog (sizes, weights, images)
3	collections	Product groupings / editions
4	orders	Paid fulfillment orders only
5	pendingcheckouts	Temporary Razorpay drafts (auto-expire)
6	reviews	Storefront testimonials
7	marketingsettings	Floating videos + WhatsApp statuses
8	media	Binary media stored in MongoDB
9	storebills	Walk-in / physical shop sales

2. Checkout data flow



Rule: A real orders document is created only after Razorpay payment signature verification. Unpaid attempts stay in pendingcheckouts and expire via TTL.

3. Collection schemas

3.1 users

Customer / admin accounts. Password hashed with bcrypt before save.

Field	Type	Required	Notes
name	String	Yes	Full name
email	String	Yes	Unique, lowercase
password	String	Yes	Hashed
phone	String	—	Unique when non-empty (partial index)
role	Enum	—	customer admin
addresses[]	Embedded[]	—	Address book
createdAt / updatedAt	Date	auto	timestamps

Embedded: addresses[]

Field	Type	Required	Notes
label	String	—	Home / Work / Other
name	String	Yes	Receiver
phone	String	Yes	Delivery phone
addressLine1	String	Yes	
addressLine2	String	—	
city / state / pincode	String	Yes	6-digit PIN
isDefault	Boolean	—	Default delivery address

3.2 products

Field	Type	Required	Notes
id	String	Yes	Public product id (unique)
name	String	Yes	
price	Number	Yes	INR
compareAt	Number	—	MRP / strike price
collection	String	Yes	→ collections.id
category	String	Yes	
sizes[]	Embedded[]	—	id, label, price
weights[]	Embedded[]	—	id, from, to, label
images[]	String[]	—	URLs / media paths
features[]	String[]	—	
topSelling / mostLoved / inStock	Boolean	—	Flags
description / tagline / badge / willow / madeIn	String	—	Merchandising

3.3 collections

Field	Type	Required	Notes
id	String	Yes	Unique business id
name	String	Yes	Display name
slug	String	Yes	URL slug (unique)
family	String	—	hard-tennis soft-tennis
familyLabel / blurb / badge / variant	String	—	
featured	Boolean	—	
sortOrder	Number	—	Default 100

3.4 orders

Real shop orders — created only after successful payment verification.

Field	Type	Required	Notes
orderId	String	Yes	Public id (unique)
status	Enum	—	ordered → accepted → packed → shipped → delivered · cancelled
paymentStatus	Enum	—	pending paid failed refunded pending_cod
paymentMethod	Enum	Yes	razorpay upi card cod
razorpayOrderId / razorpayPaymentId	String	—	Gateway ids
paymentMeta	Mixed	—	Gateway raw meta
customer{}	Object	Yes	name, phone, email
shipping{}	Object	Yes	address lines, city, state, pincode
items[]	Embedded[]	Yes	Line snapshot at purchase
subtotal / shippingFee / discount / total	Number	Yes*	*total & subtotal required
currency	String	—	Default INR
courier{}	Object	—	name, trackingId, trackingUrl, notes
statusTimestamps{}	Object	—	Per-stage dates
statusHistory[]	Embedded[]	—	from, to, changedAt, changedBy, note

Embedded: items[]

Field	Type	Required
id, name, sizeId, sizeLabel	String	Yes
weightId, weightLabel	String	—
price, qty, lineTotal	Number	Yes

3.5 pendingcheckouts

Temporary payment drafts. TTL index deletes when `expiresAt` is reached (default +24h).

Field	Type	Required	Notes
orderId	String	Yes	Reserved public id (unique)
razorpayOrderId	String	Yes	Gateway order (unique)
amountPaise	Number	Yes	Amount in paise
customer{} / shipping{}	Object	Yes	Same shape as Order
items[]	Array	Yes	Line snapshot
subtotal / shippingFee / total	Number	Yes*	
expiresAt	Date	—	TTL expireAfterSeconds: 0

3.6 reviews — Review Management

Admin-managed testimonials. Public home shows only `status: approved`.

Field	Type	Required	Notes
name	String	Yes	Reviewer name
text	String	Yes	Review body
rating	Number	—	1–5 · default 5
location	String	—	City / region
image	String	—	Optional photo URL
productId / productName	String	—	Optional product link
status	Enum	—	pending \ approved \ hidden
featured	Boolean	—	Pin on home marquee
sortOrder	Number	—	Lower = earlier
source	Enum	—	admin \ seed \ customer

Admin: GET/POST/PUT/DELETE `/api/admin/reviews` · **Public:** GET `/api/reviews` (approved only)

3.7 marketingsettings

Singleton-style settings (key: "default").

Field	Type	Required	Notes
key	String	Yes	Unique · default "default"
floatingVideos[]	Embedded[]	—	id, title, videoUrl, productPath, active, sortOrder...
whatsappStatuses[]	Embedded[]	—	mediaUrl, mediaType image video, expiresAt, CTA...

3.8 media

Field	Type	Required	Notes
filename	String	—	Original name
contentType	String	Yes	MIME type
size	Number	—	Bytes
data	Buffer	Yes	File bytes

3.9 storebills

Field	Type	Required	Notes
billId	String	Yes	Unique bill number
customerName / customerPhone	String	—	
productId / itemName	String	*	itemName required
size / weight labels	String	—	
qty / unitPrice / discount / amount	Number	*	amount required
paymentMethod	Enum	—	cash upi card
soldAt	Date	—	Sale timestamp
notes	String	—	

4. Logical relationships

From	To	How
products.collection	collections.id	Soft string reference
orders.items[].id	products.id	Snapshot at purchase
pendingcheckouts	orders	Converted after payment verify
marketing / product URLs	media._id	Via /api/media/:id
users.addresses	checkout shipping	Copied into order at pay time

MongoDB does not enforce foreign keys. Integrity is handled in application code.

5. Indexes summary

Collection	Index	Type
users	email · phone	unique · unique (partial)
products	id	unique
collections	id · slug	unique
orders	orderId · razorpayOrderId	unique · index
pendingcheckouts	orderId · razorpayOrderId · expiresAt	unique · TTL
marketingsettings	key	unique
storebills	billId · soldAt · paymentMethod	unique · indexes

6. Document shape map

users

- └─ addresses[]

products

- └─ sizes[]
- └─ weights[]

orders

- └─ customer {}
- └─ shipping {}
- └─ items[]
- └─ courier {}
- └─ statusTimestamps {}
- └─ statusHistory[]

pendingcheckouts

- └─ customer {}
- └─ shipping {}
- └─ items[]

marketingsettings

- └─ floatingVideos[]
- └─ whatsappStatuses[]

collections · reviews · media · storebills